2026 UCARE Proposal: Explainable Deep Learning using Persistent Laplacian

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1 Statement of Purpose

Deep Learning (DL) has fundamentally changed our daily lives, ranging from autonomous driving and large language models (LLMs) to embodied Artificial Intelligence (AI). However, DL models are considered "black boxes", which are obtained based on data-hungry training and deeply stacked layers. It is still unclear what the fundamental principles of DL models are, e.g., what DL models have learned in the latent layers. Existing research has shown that DL models learn to simplify the input data's topological complexity. In [1], comprehensive simulations were performed to verify that the input data's topological complexity is gradually simplified through DL models, while shallow neural networks cannot simplify the topological complexity. In [1], the persistent homology was used to evaluate the topological complexity. Topology is one of the most abstract tools in mathematics. As a result, it can only extract high-level information without geometric details. As a result, *persistent homology alone may not capture subtle geometric differences between datasets*, which limits its usefulness for explainability. **This research explores the use of Persistent Laplacian (PL), another tool from TDA, to enhance the interpretability of DL models**. PL is a newly developed topological tool [2], which captures both topological and geometric structures across multiple scales in a stable and differentiable form and has been successfully applied in fields such as computational biology, network analysis, materials science, and signal processing, where understanding global data structure is critical [3]. The primary goal of this project is to integrate PL representations into DL frameworks to analyze and explain how models learn from data, ultimately contributing to more transparent, stable, and trustworthy DL systems. This project will provide more insightful understanding of DL compared with the results in [1], while are solely based on topological information using persistent homology.

2 Research Question

The primary research question of this project is: **How DL models process input data in both topological and geometric space?** It is often assumed that there is an underlying manifold structure of the input data samples for DL research. The manifold structure can be described using topological and geometric tools. DL models try to learn effective representation of the manifold structure for classification and prediction tasks. Based on previous research [1], our hypothesis is that DL models can simplify both topological and geometric structure of the input data. As persistent homology can only extract topological information, we will use PL as the tool to answer this questions.

To address this question, the study considers several related sub-questions. First, how does PL features and their complexity change at different layers of a deep neural network? We anticipate a clear reduction of the PL feature complexity so that DL models can effectively learn patterns in the input data. Second, what is the effective PL feature to describe the topological and geometrical complexity reduction? PL provides many features with different orders and a succinct and informative one will be useful. Last, how can PL features be used to improve DL efficiency in model size and accuracy? If we can understand how DL works, we should be able to optimize it. If the PL can discover the effective operation mechanism in DL models, we will further enhance the DL model performance.

3 Significance of Research

Despite the remarkable success of deep learning in a wide range of applications, these models remain difficult to interpret due to their black-box nature. Most existing explainability methods focus on local feature importance or post-hoc numerical analysis, which provide limited insight into the global geometric and topological structure of learned representations. This gap makes it challenging to understand how deep learning models internally organize and simplify data across layers.

PL can generalize graph Laplacians to multiscale simplicial complexes: zero eigenvalues recover persistent Betti numbers, while nonzero eigenvalues capture additional geometric structure beyond what standard persistent homology can provide [3]. By enabling scalable computation of these richer spectral descriptors and integrating them as

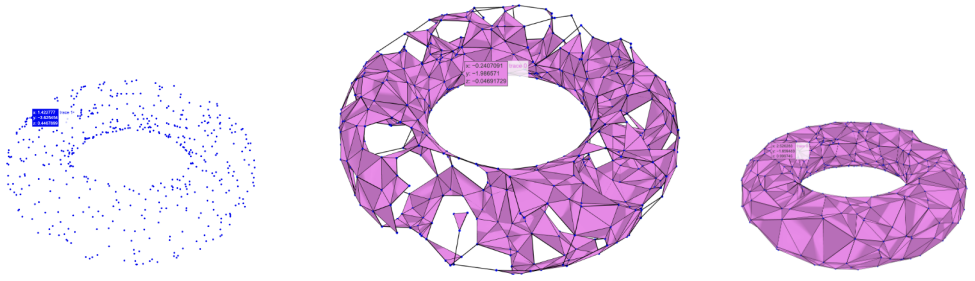


Figure 1: TDA visualizations of Persistent Laplacian.

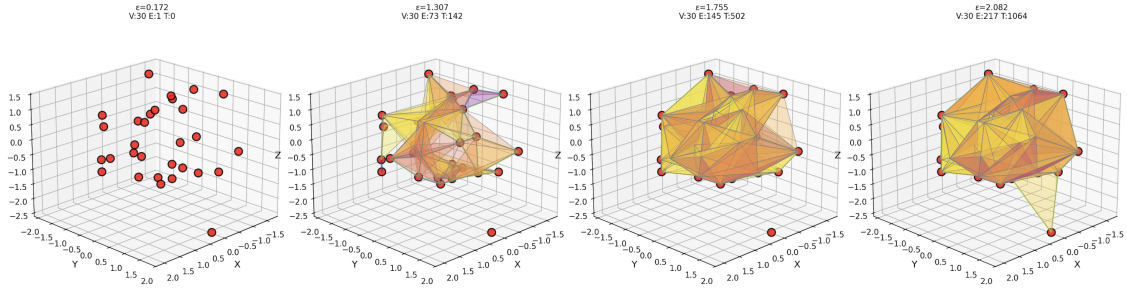


Figure 2: TDA visualizations of Persistent Homology.

regularization terms, PLs offer new insights into latent space topology.

I have been working with Dr. Guo for since Fall 2025 and learning TDA. Preliminary visualizations of simplicial complexes in Figures 1 and 2 are generated using PL library PETLS. It suggest that PL preserve important global topological structures—such as loops and higher-dimensional features—more effectively than persistent homology across multiple filtration scales. In addition, the spectral evolution of PL is smoother and more stable as connectivity increases, indicating improved robustness to combinatorial noise. These properties suggest that PL provide richer and more reliable descriptors of latent-space structure.

Figures 1 and 2 illustrate two different ways of understanding the underlying structure within data. Figure 2 introduces persistent homology, a traditional tool that tracks how topological features such as connected components, loops, and higher-dimensional holes appear and disappear as the filtration scale changes [1]. This approach provides a clear summary of the overall topological structure of the data, but it is primarily qualitative and overlooks finer geometric details. Figure 1 then expands on this idea by presenting persistent Laplacian, which incorporates spectral information through eigenvalues. In doing so, it not only shows whether topological features exist, but also reveals their geometric strength and stability across different scales.

By enabling direct integration into neural network training through loss functions or regularization terms, PL allows topology and geometry to actively shape the learning process rather than serving as a purely post-hoc analysis tool. This capability has the potential to improve model robustness, generalization, and interpretability across applications such as computer vision, signal processing, medical imaging, and robotics. Overall, this research bridges the gap between topological data analysis and practical DL models, contributing to the development of more transparent, stable, and trustworthy DL systems.

My interest in this research topic grew naturally from my plans for the future. Next semester, I will take a deep learning course, which will be my first formal step into studying advanced neural networks in depth. As I began preparing for that course, I realized I wanted to understand not only how deep learning models work, but also why they behave the way they do. This project allows me to explore that curiosity early by studying how deep learning models learn and organize information. Working on explainable deep learning has also helped me see research as a long-term path rather than a single project. Looking ahead, I plan to continue my education through a PhD program and focus my studies on deep learning, and this research experience represents an important first chapter in that journey.

4 Methods of Data Collection

The data collection process for this research follows established practices in deep learning and topological data analysis, with Persistent Laplacian (PL) construction guided by the PETLS framework [2]. The objective of data collection

is to obtain latent representations from trained neural networks and transform them into topological inputs suitable for PL computation.

The data collection process consists of the following steps:

1. **Dataset Selection and Preparation:** This study uses the publicly available MNIST dataset, which contains 70,000 grayscale images of handwritten digits from 0 to 9, divided into standard training and test sets. MNIST is a well-known benchmark in deep learning, and its clear geometric structure makes it well suited for topological analysis of both image data and latent representations. In addition to MNIST, this study also uses benchmark datasets from the MoleculeNet collection, specifically QM7 and QM7b. These datasets include molecular structures along with their quantum-chemical properties, represented using both 3D atomic coordinates and Coulomb matrices. Using these datasets allows the evaluation of topological and spectral embeddings for regression tasks involving molecular properties.

2. **Neural Network Training and Latent Data Collection:** Deep neural networks are trained on the MNIST and MoleculeNet dataset using standard optimization procedures. During training, latent representations are collected from selected hidden layers of the network. These layer-wise activations represent how the model organizes input data internally and serve as the primary source of data for Persistent Laplacian construction.

3. **Point Cloud Formation and Distance Computation:** The collected latent representations are treated as point clouds in high-dimensional space, where each point corresponds to the latent embedding of an input sample. Pairwise distance matrices are computed for these point clouds to support topological construction.

4. **Filtered Simplicial Complex Construction:** Using the computed distance matrices, filtered simplicial complexes are constructed following the PETLS pipeline [2]. Alpha complexes are used as the primary construction, with Vietoris–Rips complexes considered for comparison. Filtrations are defined across multiple scales to capture multiscale topological structure.

5. **Persistent Laplacian Data Collection:** Boundary matrices are extracted from the filtered simplicial complexes at different filtration values, and Persistent Laplacian operators are constructed. The eigenvalue spectra of these operators, computed across filtration intervals and homological dimensions, are collected as the primary dataset for this research.

The data collected for this study consist of layer-wise latent representations derived from MNIST-trained neural networks and their associated PL spectra. All data are generated during model training and topological construction, ensuring reproducibility and controlled evaluation of topological structure in deep learning representations.

5 Analysis of Data

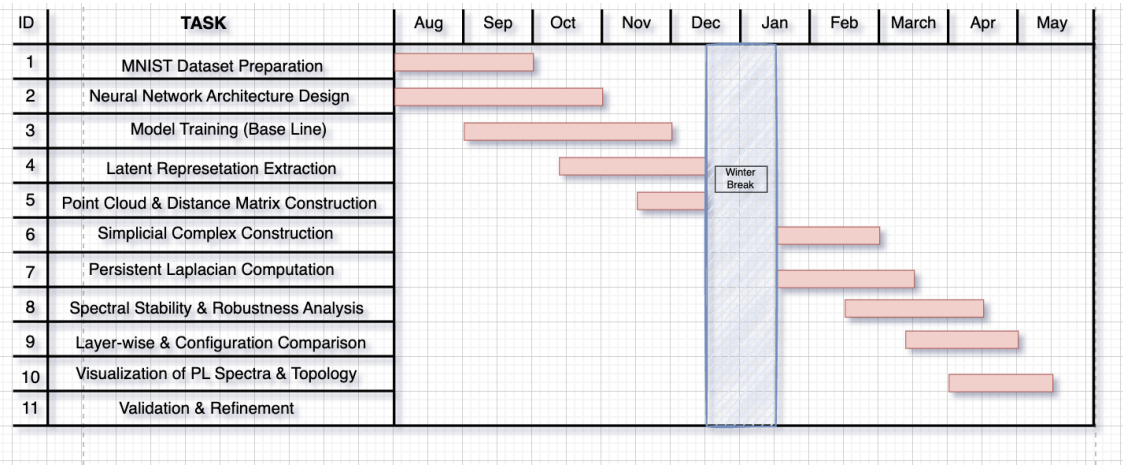


Figure 3: Research Workflow for Persistent Laplacian Analysis

The analysis focuses on understanding how deep neural networks transform and organize data in their latent spaces and how PL–based representations capture the evolution of topological and geometric structure across network layers.

Using latent representations generated from MNIST-trained models, this study analyzes PL spectra to characterize structural simplification and stability in DL models.

The following analysis procedures will be performed:

1. **Layer-wise Latent Space Analysis:** Latent representations collected from different hidden layers of the neural network are analyzed separately. Each layer's representations are treated as point clouds to study how their structure changes as depth increases.

2. **Persistent Laplacian Spectrum Computation:** For each layer-wise point cloud, PL operators are computed across multiple filtration scales following the PETLS framework [2]. Both harmonic (zero eigenvalue) and non-harmonic spectral components are examined to capture topological and geometric features.

3. **Topological Complexity Evaluation:** Changes in PL spectral properties, such as the number of near-zero eigenvalues, spectral gaps, and eigenvalue distributions, are analyzed to quantify the reduction of topological and geometric complexity across network layers.

4. **Stability and Robustness Analysis:** The stability of PL representations is evaluated by examining how PL spectra change under small perturbations, including variations in filtration parameters and sampling of latent points. This analysis assesses the robustness of PL-based descriptors.

5. **Comparative Analysis Across Training Configurations:** PL spectra are compared across different network depths and training configurations to identify consistent structural patterns. These comparisons help reveal how architectural and training choices influence latent-space topology.

To ensure the credibility and success of the proposal, the following timeline and resources will be allocated:

- **Timeline:** This project, scheduled from August 2026 to May 2027 with a commitment of around 20 hours per week, will focus on dataset preparation, model training, topological analysis, and performance evaluation. The details are given in Fig. 3.

- **Resources:** The research will utilize the laboratory's existing computational resources, including GPUs for deep learning, Python and Jupyter Notebook, relevant datasets, and libraries for topological data analysis.

6 Benchmarks

To ensure the successful completion of the project, the following benchmarks will be established to compare with the results in [1] which is solely based on topological information using Persistent Homology.

1. **Latent-Space Topology Preservation:** The benchmark involves evaluating how Persistent Laplacian–based regularization shapes the topological and geometric structure of latent representations across different network layers, ensuring meaningful global patterns are preserved.

2. **Model Performance and Stability:** Another benchmark is to maintain or improve predictive performance while enhancing stability and robustness of the learned representations. This includes assessing sensitivity to noise, perturbations, and variations in input data.

3. **Visualization and Interpretability:** The research will produce clear visualizations of latent-space structures, including spectral plots and topological summaries (e.g., Betti numbers, spectral gaps), to provide interpretable insights into how the model organizes information.

References

[1] G. Naitzat, A. Zhitnikov, and L.-H. Lim, "Topology of deep neural networks," *Journal of Machine Learning Research*, vol. 21, no. 184, pp. 1–40, 2020.

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[3] F. Mémoli, Z. Wan, and Y. Wang, "Persistent laplacians: Properties, algorithms and implications," *SIAM Journal on Mathematics of Data Science*, vol. 4, no. 2, pp. 858–884, 2022.